09. Acceptance-Rejection

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Intuition: • I want to generate r.v.s from some pdf $f(x)$

• But it is hard to do that!

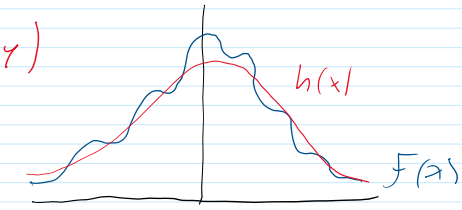
• Pick a different pdf $h(x)$

→ sample X from $h(x)$

→ ask whether or not (Binary) X looks like it came from $f(x)$

→ if so, keep X (Accept)

if not, start over (Reject)



• Something called "Rejection Sampling"

• These Accept/Reject steps occur in more complex algorithms, e.g. Metropolis Hasting, MCMC

Conditional Probability

$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ are the values of two 6-sided dice

$X_1 \in \{1, \dots, 6\}$, $X_2 \in \{1, \dots, 6\}$

But we could condition on $X_1 + X_2 = 7$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \mid \{X_1 + X_2 = 7\}$$

• Bayes Rule / Formula: $\underbrace{P(A \mid B)}_{\text{Conditional}} \underbrace{P(B)}_{\text{Marginal}} = \underbrace{P(A, B)}_{\text{Joint}}$

• Conditional Density Function:

For r.v.s X, Y, $\underbrace{f_{Y|X}(y \mid x)}_{\text{Conditional}} = \dfrac{\overbrace{f(x, y)}^{\text{Joint}}}{\underbrace{f_X(x)}_{\text{Marginal}}}$

Theorem (Von Neumann, 1951) ← But we can consider unbounded intervals

Let $f(x)$ be a pdf on $[a, b]$ and $h(x)$ another pdf on $[a, b]$.

We write $f(x) = c\, g(x)\, h(x)$ → $\dfrac{f(x)}{h(x)} = c\, g(x)$

Scale ↓ $[0,1]$ ↙

where • $h(x) \geq 0$, $\int_a^b h(x)\, dx = 1$ (pdf)

• $g(x) \in [0, 1]$ ← any function with values in $[0,1]$

pick Z from $h()$

- $g(x) \in [0,1]$ ← any function with values in $[0,1]$
- $C \geq 1$

If $Z \sim h(z)$, $U \sim Unif[0,1]$, $U \perp\!\!\!\perp Z$ Independent

Then $Z \mid \{U \leq g(Z)\} \sim f(x)$

$[0,1]$ $[0,1]$

Accept / Reject, i.e. keep Z if $U \leq g(Z)$

Proof: Start joint pdf of U, Z

$$f_{UZ}(u,z) = f_U(u)\, f_Z(z) = \mathbb{1}[u \in [0,1]]\, h(z)$$

Joint — prod. Marginal due to independence

Bayes: $P(\{Z \leq z\} \mid \{U \leq g(Z)\})\, P(\{U \leq g(Z)\}) = P(\{Z \leq z\}, \{U \leq g(Z)\})$

$P(A \mid B)\, P(B) = P(A, B)$

- $P(\{Z \leq z\}, \{U \leq g(Z)\}) = \int_0^z \int_0^{g(z)} f_{UZ}(u,z)\, du\, dz$

Take a derivative w.r.t. z to get conditional pdf

$$h(z \mid U \leq g(Z))\, P(U \leq g(Z)) = \int_0^{g(z)} f_{UZ}(u,z)\, du\, dz$$

$\mathbb{1}[u \in [0,1]]\, h(z)$

$$h(z \mid U \leq g(Z)) = \frac{\int_0^{g(z)} f_{U,Z}(u,z)\, du\, dz}{P(U \leq g(Z))}$$

$\frac{1}{C}$

Top $= g(z)\, h(z)$

Bottom $= \iint_{[0,1]\times[a,b]} \mathbb{1}[u \leq g(z)]\, du\; h(z)\, dz = \int_a^b g(z)\, h(z)\, dz$

we choose $C = \left[\int_a^b g(z)\, h(z)\, dz\right]^{-1}$

Then $h(z \mid U \leq g(z)) = \dfrac{g(z)\, h(z)}{1/C} = C\, g(z)\, h(z) = f(z)$

$\square$